

Hirsch Chheda

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EDUCATION

University of California-Riverside

B.S. in Computer Engineering | GPA: 3.7/4.0

March 2025

Relevant Courses: Data Structures, Natural Language Processing, Algorithms & Analysis, AI, Machine Learning, Web Dev

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, SQL, Assembly, Swift, MATLAB

Database: AWS DynamoDB, Oracle, Microsoft SQL Server, MySQL, PostgreSQL, MongoDB

Web Development: HTML, CSS, JavaScript, jQuery, React.js, Node.js, Figma

Tools: AWS, Git, Visual Studio Code, Jupyter, CLion

Libraries: Pandas, NumPy, Tensorflow, Scikit-Learn, Keras, Natural Language Toolkit

INTERNSHIP EXPERIENCE

Data Automation Intern | Kyobee | Cerritos, CA

June 2024 - September 2024

- Collaborated with a team using automation tools and **SQL** algorithms, improving 94,000 database entries' accuracy and query performance by 30%.
- Developed automation scripts with SQL and testing automation, reducing manual data entry errors by 40% and enhancing data consistency.
- Executed automated tests, debugged scripts, and ensured seamless system integration with accurate data visualization for better decision-making.

Software Engineer Intern | First American | Santa Ana, CA

June 2023 - August 2023

- Invested 200+ hours mastering **JavaScript**, **HTML**, **CSS**, and **React.js**, creating responsive web apps that boosted user engagement and enhanced client-side experiences.
- Developed unit and integration tests for WireLink UI using React.js, improving system reliability and supporting \$5M+ in daily transactions utilizing **VS Code**.
- Gained proficiency in **API** lifecycle management, optimizing data flow between front-end and server.
- Utilized **Cypress** to automate regression and smoke tests for React.js apps, cutting pre-release bugs by 60% and minimizing transaction-related downtime.

WORK EXPERIENCE

Lab Assistant | UCR, Center For Environmental Research And Technology

May 2023 - March 2024

- Actively engaged in groundbreaking research initiatives that focused on innovating vehicle technologies to enhance fuel efficiency and reduce emissions.
- With the help of PHD students developed prototypes that demonstrated a potential reduction in carbon emissions by up to 20% in preliminary tests.
- Managed the calibration, maintenance, and installing of the OSAR (Onboard Sensing, Analysis and Reporting) testing apparatus resulting in improved data precision and operational efficiency.

PROJECTS

Finance Tracker | Personal Project | <https://github.com/Hirschheda/Finance-Tracker>

- Developed a full-stack Finance Tracker web application using **React.js** for the frontend and **AWS Lambda** (serverless) with **API Gateway** for backend transaction management.
- Implemented secure user authentication and authorization with **Amazon Cognito** and **OIDC**, enabling users to manage personal financial transactions and visualize spending with interactive charts (Recharts).
- Integrated **Axios** for RESTful API communication, and utilized Git and GitHub for version control and collaborative development.

S&P500 Forecast | Personal Project | <https://github.com/Hirschheda/sp500-forecast>

- Automated data ingestion from Yahoo Finance, resampled to Friday-close bars, and engineered 10+ technical and macro features to feed into supervised models.
- Trained **LightGBM** regressors and **XGBoost** classifiers with hold-out validation for next-week price and direction, and developed a Keras **LSTM** with **Tensorflow** and **KerasTuner** hyperparameter search.

IKEA Rogue-Like Game | Group Project | <https://github.com/Hirschheda/IKEA-GAME>

- Collaborated with a team of 5 others and developed a game where the main character is stuck in an IKEA and has to escape while fighting off pieces of furniture.
- Created the player movement/dash mechanism, game sounds, and room designs with **Python** through **GoDot**, and open-source game engine